Unramifiedness of $\mathcal{P}^{[d]_G}$ at the exceptional divisor

(standalone excerpt)

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May 27, 2026

Roadmap. Section 1 collects, without re-deriving them, the facts about symmetric powers of torsors that the proof uses, stated over an abstract base. Section 2 specialises this to a smooth projective curve: it recalls the notion of bounded ramification and fixes the geometric data (the symmetric tower over $U^{(d)}$, the compactification $C' \to C$, the blowups at the points at infinity). Section 3 assembles everything into a single master diagram, and Section 4 states and proves the unramifiedness theorem.

1 Background

Throughout this section S is a base scheme, G a finite abelian group, $H \subseteq G$ a subgroup, and $d \geq 0$ an integer. All bases are assumed Zariski-locally quasi-projective over S, so that the symmetric-power quotients below exist as schemes.

1.1 Symmetric powers of torsors

Let $\mathcal{P}$ be a G-torsor over an S-scheme X. For each $i \in [\![1, d]\!]$ write $p_i : X^{\times_S d} \to X$ for the i-th projection and consider the G-torsor

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} \;=\; K_G \backslash \mathcal{P}^{\times_S d}$$

over $X^{\times_S d}$, where $K_G \subseteq G^d$ is the kernel of the multiplication morphism $G^d \to G$ (the dependence on d is suppressed in the notation). The symmetric group S_d acts on the right on $K_G \backslash \mathcal{P}^{\times_S d}$ by $(p_i)_i \cdot \sigma = (p_{\sigma(i)})_i$, and this action commutes with the left G-action.

Proposition 1 ([Gui19], Proposition 2.32)**.** *The right action of S_d on $K_G \backslash \mathcal{P}^{\times_S d}$ is admissible, so that the quotient $\mathcal{P}^{[d]}$ of $K_G \backslash \mathcal{P}^{\times_S d}$ by S_d exists as a scheme over S. Moreover, the canonical morphism $\mathcal{P}^{[d]} \to \mathrm{Sym}^d_S(X)$ is a G-torsor, and the morphism*

$$p_1^{-1}\mathcal{P} \otimes^G \cdots \otimes^G p_d^{-1}\mathcal{P} \to r^{-1}\mathcal{P}^{[d]},$$

where $r : X^{\times_S d} \to \mathrm{Sym}^d_S(X)$ is the canonical projection, is an isomorphism of G-torsors over $X^{\times_S d}$.

When G is abelian the condition $\prod_{i=1}^d g_i = 1$ defining K_G is invariant under permuting the coordinates, so the (left) S_d-action normalises the K_G-action. Writing $\Xi = K_G \rtimes S_d$ for the resulting finite group, acting on $\mathcal{P}^{\times_S d}$ by

$$\big((g_1,\dots,g_d),\sigma\big) \star (p_1,\dots,p_d) = \big(g_1 p_{\sigma^{-1}(1)},\ \dots,\ g_d p_{\sigma^{-1}(d)}\big),$$

one has $(K_G\backslash\mathcal{P}^{\times_S d})/S_d \cong \Xi\backslash\mathcal{P}^{\times_S d}$.

Corollary 2. *There is a natural canonical morphism*

$$q : S_d\backslash\mathcal{P}^{\times_S d} \longrightarrow \Xi\backslash\mathcal{P}^{\times_S d} \qquad \textit{equivalently} \qquad q : S_d\backslash\mathcal{P}^{\times_S d} \longrightarrow (K_G\backslash\mathcal{P}^{\times_S d})/S_d,$$

the comparison between the symmetric power of $\mathcal{P}$ as a scheme *and the symmetric power of $\mathcal{P}$* as a G-torsor. *We write $\mathcal{P}^{(d)}$ for the former and $\mathcal{P}^{[d]}$ for the latter, so that $q : \mathcal{P}^{(d)} \to \mathcal{P}^{[d]}$.*

Proof. S_d is a subgroup of Ξ, hence any Ξ-invariant morphism is automatically S_d-invariant. □

1.2 The canonical decomposition of a torsor

Proposition 3. *Let $\mathcal{P} \to X$ be a G-torsor in $X_{ét}$ and let $H \subseteq G$ be a subgroup, with quotient map $\pi : G \to G/H$. Then $\mathcal{P} \to X$ factors*

$$\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X,$$

uniquely up to a unique isomorphism commuting with the projections, where

1. *$\mathcal{P}_{G/H} := \mathcal{P} \times^G (G/H)$ is a G/H-torsor over X, and*
2. *$\mathcal{P}$ is an H-torsor over the scheme $\mathcal{P}_{G/H}$ via ϕ.*

1.3 The tower of symmetric quotients

Fix a G-torsor $\mathcal{P} \to X$ and a subgroup $H \subseteq G$, with canonical factorisation $\mathcal{P} \xrightarrow{\phi} \mathcal{P}_{G/H} \xrightarrow{\psi} X$ as in Proposition 3 (ϕ an H-torsor, ψ a G/H-torsor). We record how the symmetric-power construction of Section 1.1 interacts with this factorisation.

Summation kernels. Besides $K_G = \ker(G^d \to G)$ we need

$$K_H := \ker(H^d \to H), \qquad K_{G/H} := \ker\big((G/H)^d \to G/H\big).$$

The snake lemma applied to

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^d & \longrightarrow & G^d & \longrightarrow & (G/H)^d & \longrightarrow & 0 \\ & & \downarrow\!\!\!\!\downarrow & & \downarrow\!\!\!\!\downarrow & & \downarrow\!\!\!\!\downarrow & & \\ 0 & \longrightarrow & H & \longrightarrow & G & \longrightarrow & G/H & \longrightarrow & 0, \end{array}$$

(the vertical maps being the surjective summation morphisms) gives a short exact sequence

$$0 \to K_H \to K_G \to K_{G/H} \to 0, \tag{1}$$

hence $K_G/K_H \cong K_{G/H}$. Inside G^d we have the chains of subgroups

$$K_H \ \subseteq\ K_G \ \subseteq\ H^d + K_G, \qquad H^d \ \subseteq\ H^d + K_G.$$

Four quotient schemes. Let S_d act on $\mathcal{P}^{\times_S d}$ as in Corollary 2, and let each of the subgroups above act diagonally via the G-action on $\mathcal{P}$. Set

$$
\begin{aligned}
\mathcal{P}^{[d]_G} &:= (K_G \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, && \text{the symmetric } G\text{-power of } \mathcal{P};\\
\mathcal{P}^{[d]_H} &:= (K_H \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, && \text{the symmetric } H\text{-power of } \mathcal{P} \text{ relative to } \phi;\\
(\mathcal{P}_{G/H})^{(d)} &:= (H^d \rtimes S_d)\backslash \mathcal{P}^{\times_S d}, && \text{the symmetric power of the scheme } \mathcal{P}_{G/H};\\
(\mathcal{P}_{G/H})^{[d]} &:= \big((H^d + K_G) \rtimes S_d\big)\backslash \mathcal{P}^{\times_S d}, && \text{the symmetric } G/H\text{-power of } \mathcal{P}_{G/H}.
\end{aligned}
$$

Proposition 4. *The four quotient schemes above exist and fit into a canonical commutative diagram*

$$
\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}_{G/H})^{(d)} \arrow[d, "q"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}_{G/H})^{[d]} \arrow[d, "G/H"] \\
& X^{(d)},
\end{tikzcd}
$$

in which each labelled arrow is a torsor under the indicated constant group, the right-hand vertical q is the canonical projection of Corollary 2 applied to $\mathcal{P}_{G/H} \to X$, and the upper square is Cartesian:

$$\mathcal{P}^{[d]_H} \cong \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}.$$

The unlabelled left-hand vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is then the base change of q along $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$. Neither q nor this left-hand vertical is, in general, a torsor under a constant group scheme.

Proof. The four labelled arrows are torsors under the indicated constant groups by applying Proposition 1 and Proposition 3:

- to the G-torsor $\mathcal{P} \to X$, giving the G-torsor $\mathcal{P}^{[d]_G} \to X^{(d)}$ (the diagonal arrow);
- to the H-torsor $\phi : \mathcal{P} \to \mathcal{P}_{G/H}$ (with $\mathcal{P}_{G/H}$ as base), giving the H-torsor $\mathcal{P}^{[d]_H} \to (\mathcal{P}_{G/H})^{(d)}$ (the top arrow);
- to the G/H-torsor $\psi : \mathcal{P}_{G/H} \to X$, giving the G/H-torsor $(\mathcal{P}_{G/H})^{[d]} \to X^{(d)}$ (the bottom-right vertical) and, via Corollary 2, the comparison map $q : (\mathcal{P}_{G/H})^{(d)} \to (\mathcal{P}_{G/H})^{[d]}$.

The factorisation $\mathcal{P}^{[d]_G} \xrightarrow{H} (\mathcal{P}_{G/H})^{[d]} \xrightarrow{G/H} X^{(d)}$ of the diagonal arrow is the canonical decomposition of Proposition 3 applied to $\mathcal{P}^{[d]_G} \to X^{(d)}$ and $H \subseteq G$: both $\mathcal{P}^{[d]_G} \times^G (G/H)$ and $(\mathcal{P}_{G/H})^{[d]}$ are realised as the admissible quotient $\big((H^d + K_G) \rtimes S_d\big)\backslash \mathcal{P}^{\times_S d}$.

For the Cartesian property, set $F := \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)}$. Base-changing the H-torsor $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$ along q exhibits F as an H-torsor over $(\mathcal{P}_{G/H})^{(d)}$. The universal property of the fibre product gives a canonical H-equivariant morphism $\mathcal{P}^{[d]_H} \to F$ over $(\mathcal{P}_{G/H})^{(d)}$; any morphism of H-torsors is an isomorphism, so $\mathcal{P}^{[d]_H} \cong F$ and the upper square is Cartesian. Via this square the left vertical $\mathcal{P}^{[d]_H} \to \mathcal{P}^{[d]_G}$ is identified with the base change of q along $\mathcal{P}^{[d]_G} \to (\mathcal{P}_{G/H})^{[d]}$. $\square$

2 The geometric setting

2.1 Bounded ramification along a curve

Let C be a smooth projective geometrically connected curve over a field k, let $\mathfrak{m}$ be a fixed modulus, and put $U = C \setminus \mathfrak{m}$. Let $\mathcal{P} \to U$ be a G-torsor with G finite abelian. For a closed point P of C, passing to the completion $\widehat{\mathcal{O}}_{C,P}$ gives the complete local field $\widehat{K}_P$, and restricting $\mathcal{P}$ to $\mathrm{Spec}(\widehat{K}_P)$ yields a G-torsor over a field, hence a disjoint union of spectra of finite separable field extensions

$$\mathcal{P}|_{\mathrm{Spec}(\widehat{K}_P)} \cong \bigsqcup \mathrm{Spec}(F),$$

in which the extensions $F/\widehat{K}_P$ are all isomorphic and Galois, $H = \mathrm{Gal}(F/\widehat{K}_P)$ is the stabiliser of a connected component (a subgroup of G), and there are $[G:H]$ components. Write H^r for the r-th higher ramification group of $F/\widehat{K}_P$.

Definition 5. The extension $F/\widehat{K}_P$ *has ramification bounded by* r if the ramification group H^r is trivial. We say $\mathcal{P}$ *has ramification bounded by* r *at* P if the corresponding local extensions $F/\widehat{K}_P$ satisfy this condition.

2.2 The symmetric tower over $U^{(d)}$

For the rest of the note we fix, in addition to the data above:

- a normal subgroup $H \trianglelefteq G$;
- a point $p_\infty \in \mathfrak{m}$ and an integer $d \geq 1$;

and we assume that $\mathcal{P} \to U$ has ramification bounded by d at p_∞ in the sense of Definition 5 (i.e. the d-th higher ramification group of the extension of DVRs at p_∞ is trivial).

Apply the construction of Section 1.3 with $S = \mathrm{Spec}\, k$ and base $X = U$. This produces the four quotient schemes

$$\mathcal{P}^{[d]_H}, \qquad \mathcal{P}^{[d]_G}, \qquad (\mathcal{P}_{G/H})^{(d)}, \qquad (\mathcal{P}_{G/H})^{[d]},$$

and, by Proposition 4, the canonical commutative diagram with Cartesian upper square

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}_{G/H})^{(d)} \arrow[d, "q"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}_{G/H})^{[d]} \arrow[d, "G/H"] \\
& U^{(d)},
\end{tikzcd}$$

in which the arrows labelled H, G/H and G are torsors under the indicated finite constant groups (in particular, étale).

2.3 Compactification and blowup

The G/H-torsor $\mathcal{P}_{G/H} \to U$ compactifies to a finite morphism of smooth projective curves $f\colon C' \to C$; fix $p'_\infty \in f^{-1}(p_\infty)$. Taking d-fold symmetric powers gives the finite morphism

$$f^{(d)}\colon C'^{(d)} \to C^{(d)}, \qquad d \cdot p'_\infty \longmapsto d \cdot p_\infty.$$

Let

$$\pi_C\colon X_C \to C^{(d)}, \qquad \pi_{C'}\colon X_{C'} \to C'^{(d)}$$

be the blowups at $d \cdot p_\infty$ and $d \cdot p'_\infty$ respectively, with exceptional divisors $E_C, E_{C'}$ and generic points $\eta_C, \eta_{C'}$. The local rings at these generic points are the discrete valuation rings

$$V_C := \mathcal{O}_{X_C,\eta_C} \subset K(C^{(d)}), \qquad V_{C'} := \mathcal{O}_{X_{C'},\eta_{C'}} \subset K(C'^{(d)}).$$

3 The master diagram

The torsor tower over $U^{(d)}$ embeds into the projective geometry of $C^{(d)}$ and $C'^{(d)}$, which in turn carry the canonical blowups at the points at infinity. Assembling all of this:

$$\begin{tikzcd}
\mathcal{P}^{[d]_H} \arrow[r, "H"] \arrow[d] & (\mathcal{P}_{G/H})^{(d)} \arrow[r, hook, "\iota'"] \arrow[d, "q"] & C'^{(d)} \arrow[dd, "f^{(d)}"] & X_{C'} \arrow[l, "\pi_{C'}"'] \arrow[dd, dashed, "f_X"] \\
\mathcal{P}^{[d]_G} \arrow[r, "H"'] \arrow[dr, bend right, "G"'] & (\mathcal{P}_{G/H})^{[d]} \arrow[d, "G/H"] & & \\
& U^{(d)} \arrow[r, hook, "\iota"'] & C^{(d)} & X_C \arrow[l, "\pi_C"]
\end{tikzcd}$$

Reading the diagram from left to right:

Left block (columns 1–2): the torsor tower of Proposition 4 over $U^{(d)}$ — H-torsors horizontally, the G/H-torsor on the lower right, and the composite G-torsor $\mathcal{P}^{[d]_G} \to U^{(d)}$.

Middle block (column 3): the compactifications. Here ι is the canonical open immersion $U^{(d)} \hookrightarrow C^{(d)}$, and ι' is the open immersion $(\mathcal{P}_{G/H})^{(d)} \hookrightarrow C'^{(d)}$ coming from $\mathcal{P}_{G/H} \subset C'$. Restricting $f^{(d)}$ along ι' recovers the canonical map $(\mathcal{P}_{G/H})^{(d)} \to U^{(d)}$, i.e. the composite of q with the G/H-torsor.

Right block (column 4): the blowups at the diagonal points at infinity. The dashed arrow $f_X\colon X_{C'} \dashrightarrow X_C$ is the rational lift of $f^{(d)}$, defined on the strict transform; on generic points of the exceptional divisors it sends $\eta_{C'} \mapsto \eta_C$, inducing an extension of DVRs $V_C \hookrightarrow V_{C'}$.

4 Unramifiedness of $\mathcal{P}^{[d]_G}$ at η_C

Theorem 6. *Assume*

(H1) $(\mathcal{P}_{G/H})^{[d]} \to U^{(d)}$ *is unramified at* η_C*;*

(H2) *$\mathcal{P}^{[d]_H} \to (\mathcal{P}_{G/H})^{(d)}$ is unramified at $\eta_{C'}$;*

(H3) *$G = \mathbb{Z}/p^r\mathbb{Z}$ is cyclic of prime-power order.*

Then $\mathcal{P}^{[d]_G} \to U^{(d)}$ is unramified at η_C.

Proof. The proof has two preliminary reductions and four steps.

Preliminary reductions.

(R1) *$\mathcal{P} \to U$ is connected.* Let $\rho\colon \pi_1(U) \to G$ be the monodromy of $\mathcal{P}$ and put $G_0 := \mathrm{Im}(\rho)$, $H_0 := H \cap G_0$. Pick a connected component $\mathcal{P}_0 \subseteq \mathcal{P}$; it is a connected G_0-torsor, and $\mathcal{P}$ is the disjoint union of $[G : G_0]$ copies of $\mathcal{P}_0$ permuted by G. Since the symmetric-power quotients of Section 1.3 commute with such disjoint unions, each of the four schemes in Proposition 4 for $(\mathcal{P}, G, H)$ decomposes as a disjoint union of copies of the corresponding scheme for $(\mathcal{P}_0, G_0, H_0)$. Unramifiedness at η_C (resp. $\eta_{C'}$) is detected componentwise, so (H1) and (H2) are inherited by $(\mathcal{P}_0, G_0, H_0)$; (H3) is inherited because subgroups of cyclic p-groups are cyclic p-groups. Replacing $(\mathcal{P}, G, H)$ by $(\mathcal{P}_0, G_0, H_0)$ we may assume $\mathcal{P}$ is connected and ρ is surjective.

(R2) $H \neq 0$. If $H = 0$ then $\mathcal{P}_{G/H} = \mathcal{P}$ and $(\mathcal{P}_{G/H})^{[d]} = \mathcal{P}^{[d]_G}$, so (H1) is literally the conclusion.

Under (R1)–(R2) the function fields

$$E := K(\mathcal{P}^{[d]_G}), \qquad F := K\big((\mathcal{P}_{G/H})^{[d]}\big)$$

are single fields, Galois over $K_C := \operatorname{Frac} V_C$ with groups G and G/H respectively, and Proposition 4 yields a tower $K_C \subseteq F \subseteq E$ with $\mathrm{Gal}(E/F) = H$.

Step 1: A place V_F of F above V_C, unramified by (H1). The comparison map $q : (\mathcal{P}_{G/H})^{(d)} \to (\mathcal{P}_{G/H})^{[d]}$ is generically finite and dominant; combined with the open immersions $(\mathcal{P}_{G/H})^{(d)} \hookrightarrow C'^{(d)}$ and $(\mathcal{P}_{G/H})^{[d]} \to C^{(d)}$ (via the G/H-torsor and the open immersion $U^{(d)} \hookrightarrow C^{(d)}$), this gives $F \subseteq K_{C'}$, where $K_{C'} := \operatorname{Frac} V_{C'}$. Restriction of $V_{C'}$ to F defines a DVR V_F with $V_F \mid V_C$. Hypothesis (H1) asserts that *every* place of F above V_C is unramified; in particular V_F/V_C is unramified.

Fix a place v_E of E above V_F and, inside a chosen algebraic closure of K_C, a place $v_\star$ of the compositum $E \cdot K_{C'}$ above both v_E and $V_{C'}$. Since $H = \mathrm{Gal}(E/F)$ acts transitively on the places of E above V_F, triviality of $I_{v_E}(E/F)$ implies triviality of inertia at *every* place of E above V_F; combined with the unramified step V_F/V_C, that yields the conclusion. It therefore suffices to prove

$$I := I_{v_E}(E/F) \;=\; 1 \qquad \text{inside } H.$$

Step 2: $E \not\subseteq K_{C'}$. Let $L := K(\mathcal{P}^{\times d})$. By (R1), $\mathcal{P}^{\times d} \to U^d$ is a connected G^d-torsor; the S_d-action on $\mathcal{P}^{\times d}$ covering the S_d-action on U^d exhibits L/K_C as a Galois extension with group

$$\widetilde{G} \;:=\; G^d \rtimes S_d.$$

(Generically $L/K(U^d)$ is Galois with group G^d and $K(U^d)/K_C$ is Galois with group S_d; the two pieces form a semidirect product because the S_d-permutation of factors normalises the G^d-action.)

By the definition of $\mathcal{P}^{[d]_G}$ as $\Xi\backslash\mathcal{P}^{\times d}$ with $\Xi := K_G \rtimes S_d \trianglelefteq \widetilde{G}$ and quotient $\widetilde{G}/\Xi = G^d/K_G = G$, Galois correspondence gives $E = L^{\Xi}$. Likewise $(\mathcal{P}_{G/H})^{(d)} = (H^d \rtimes S_d)\backslash\mathcal{P}^{\times d}$ yields $K_{C'} = L^{H^d \rtimes S_d}$. Hence

$$E \subseteq K_{C'} \iff \Xi \supseteq H^d \rtimes S_d \iff K_G \supseteq H^d.$$

For any $h \in H$ the element $(h, 0, \ldots, 0) \in H^d$ has summation h in G, so $K_G \supseteq H^d$ would force $h = 0$ for every $h \in H$, contradicting (R2). Therefore $E \not\subseteq K_{C'}$.

Step 3: Inertia comparison via (H2). Set $M := E \cap K_{C'}$, so $F \subseteq M \subsetneq E$ (the inclusion $F \subseteq M$ because $F \subseteq E \cap K_{C'}$; the proper inclusion by Step 2). Then $E \cdot K_{C'}$ is Galois over $K_{C'}$, and restriction to E identifies

$$N \;:=\; \mathrm{Gal}(E \cdot K_{C'}/K_{C'}) \;\cong\; \mathrm{Gal}(E/M) \;\leq\; \mathrm{Gal}(E/F) \;=\; H, \qquad N \neq 1.$$

The Cartesian property of Proposition 4 gives

$$\mathcal{P}^{[d]_H} \;=\; \mathcal{P}^{[d]_G} \times_{(\mathcal{P}_{G/H})^{[d]}} (\mathcal{P}_{G/H})^{(d)},$$

so the generic fibre of $\mathcal{P}^{[d]_H} \to (\mathcal{P}_{G/H})^{(d)}$ over $\operatorname{Spec} K_{C'}$ is $E \otimes_F K_{C'}$. Since E/F is Galois with group H, this tensor product is a finite product of fields, each $K_{C'}$-isomorphic to $E \cdot K_{C'}$ and Galois over $K_{C'}$ with group N; the number of factors is $[H : N]$. By (H2) each factor is unramified at $V_{C'}$; in particular

$$I' \;:=\; I_{v_\star}\big(E \cdot K_{C'}/K_{C'}\big) \;=\; 1.$$

We claim that, under $N \hookrightarrow H$,

$$I' \;=\; I \cap N \qquad \text{inside } H. \tag{2}$$

This is the standard *inertia-in-compositum* identity for the Galois extensions E/F and $K_{C'}/F$ inside their compositum, valid once the residue extension $\kappa(v_E)/\kappa(V_F)$ is separable (Bourbaki, *Algèbre Commutative*, Ch. VI, §8, n.º 3; Serre, *Local Fields*, Ch. I, §7). We verify this separability unconditionally.

Separability of residue extensions is automatic. Let L/K be any finite Galois extension of henselian DVRs with Galois group Γ, ramification index $e := |I_\Gamma|$ and residue degree $f := [\kappa_L : \kappa_K]$. Henselicity gives a unique extension of the place of K to L, so the fundamental equality

$$|\Gamma| \;=\; e \cdot f \qquad \text{(Serre, \textit{Local Fields}, Ch. I, §7, Prop. 22)}$$

holds. Since I_Γ is the kernel of the natural action $\Gamma \to \mathrm{Aut}_{\kappa_K}(\kappa_L)$, the quotient Γ/I_Γ embeds in $\mathrm{Aut}_{\kappa_K}(\kappa_L)$; combining with the bound $|\mathrm{Aut}_{\kappa_K}(\kappa_L)| \leq [\kappa_L : \kappa_K]_s$ gives

$$f \;=\; |\Gamma/I_\Gamma| \;\leq\; |\mathrm{Aut}_{\kappa_K}(\kappa_L)| \;\leq\; [\kappa_L : \kappa_K]_s \;\leq\; f.$$

All inequalities collapse, forcing $f = [\kappa_L : \kappa_K]_s$, i.e. κ_L/κ_K is separable; no assumption on $\operatorname{char} k$ or perfectness of κ_K is used. Applying this (after passing to the henselisation, where the decomposition group is the full local Galois group) to E/F at the place v_E shows $\kappa(v_E)/\kappa(V_F)$ is separable; analogously $\kappa(V_{C'})/\kappa(V_F)$ is separable.

Proof of equation (2). Separability of both residue extensions gives the standard compositum identity $\kappa(v_\star) = \kappa(v_E) \cdot \kappa(V_{C'})$ (Bourbaki, *loc. cit.*). Both inclusions follow:

- If $\sigma \in I'$ then σ acts trivially on $\kappa(v_\star)$, hence on $\kappa(v_E) \subseteq \kappa(v_\star)$, so $\sigma|_E \in I$ and $\sigma \in I \cap N$.
- Conversely, if $\sigma \in I \cap N$ then σ fixes $K_{C'}$ (whence acts trivially on $\kappa(V_{C'})$) and $\sigma|_E$ acts trivially on $\kappa(v_E)$; since $\kappa(v_\star) = \kappa(v_E) \cdot \kappa(V_{C'})$, σ acts trivially on $\kappa(v_\star)$, i.e. $\sigma \in I'$.

Combining $I' = 1$ with equation (2) yields $I \cap N = 1$ inside H.

Step 4: Cyclic prime-power structure forces $I = 1$. By (H3) the group $H \leq G = \mathbb{Z}/p^r\mathbb{Z}$ is cyclic of prime-power order, so its subgroup lattice is a chain. Consequently, two subgroups $A, B \leq H$ satisfy $A \cap B = 1$ iff $A = 1$ or $B = 1$. Applied to $I \cap N = 1$ with $N \neq 1$ (Step 3), this forces $I = 1$.

Thus E/F is unramified at V_F. Combined with the unramified extension V_F/V_C of Step 1, E/K_C is unramified at V_C, i.e. $\mathcal{P}^{[d]_G} \to U^{(d)}$ is unramified at η_C. □

Remark. The three hypotheses play disjoint roles:

- (H1) is used in Step 1 to produce the unramified extension V_F/V_C and to close the tower at the end.
- (H2) is used in Step 3 to give $I' = 1$.
- (H3) is used *only* in Step 4, to convert "$I \cap N = 1$ with $N \neq 1$" into $I = 1$; for $G = (\mathbb{Z}/2)^2$ two nontrivial subgroups can intersect trivially and the conclusion can fail.

No separability hypothesis on the residue extension $\kappa(\eta_{C'})/\kappa(\eta_C)$ is needed: the henselian fundamental equality $|\Gamma| = e \cdot f$ together with $|\Gamma/I_\Gamma| \leq [\kappa_L : \kappa_K]_s$ forces separability of the Galois residue extensions $\kappa(v_E)/\kappa(V_F)$ and $\kappa(V_{C'})/\kappa(V_F)$ irrespective of perfectness or characteristic of the base.

The bounded-ramification assumption on $\mathcal{P}$ at p_∞ does not enter the argument above directly; its role is upstream, ensuring that the rational map $f_X \colon X_{C'} \dashrightarrow X_C$ sends $\eta_{C'} \mapsto \eta_C$ and so produces the finite extension of DVRs $V_C \hookrightarrow V_{C'}$ on which Steps 1 and 3 rely.

References

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